Moving Data and AI Into Practice

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Data: The Foundation of Responsible AI

- The NRC is implementing Executive Order 14179 and the NRC AI Strategic Plan to accelerate responsible AI adoption.
- AI success depends on secure, accessible, and trusted data.
- Over the past several years, NRC has advanced a comprehensive data strategy that treats data as a strategic asset.
- Strong data foundations enable:
  - Regulatory insight
  - Operational efficiency
  - New analytical capabilities

Key Message: AI is only as effective as the data behind it.

Improved Decisions • Reduced Silos • Responsible AI Enablement • Increased Transparency

Unlocking the Value of NRC Data

- The NRC generates enormous amounts of information through its regulatory mission:
  - Licensing documents
  - Inspection findings
  - Safety analyses
  - Environmental reviews
  - Decades of technical knowledge

- **Challenge**: Much of this information historically exists in:

  - Documents
  - Legacy systems
  - Unstructured records
  - Opportunity

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Building the Data Foundations

Our data strategy focuses on three key priorities

1) Structure Data

Standardized formats	Improved metadata	Modern data architecture

2) Integrate Data Across Systems

Break down program silos	Connect datasets across agency platforms	Enable enterprise analytics

3) Make Knowledge Discoverable

Data catalogs	Improved search	AI-enabled knowledge tools

OUTCOME: Move from isolated information to enterprise insights.

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Ensuring Data Is Secure, Accessible, and Trusted

For a safety regulator, trust in data is **essential**.

NRC's data ecosystem emphasizes

Secure - protecting sensitive information	Accessible - reducing barriers to data use	Trusted - strong governance, quality, and lineage	Agency-Wide Data Stewardship

Through the Data as a Service model, NRC supports

Data scientists	Data stewards	Data analysts	Data champions

Goal: Build data literacy and shared accountability across the agency.

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Turning Data into Mission Impact

Stronger data foundations enable NRC to:

- Identify trends faster
- Perform deeper analytical reviews
- Support risk-informed regulatory decisions

New capabilities include:

- Advanced analytics
- Workflow automation
- Intelligent document processing
- AI-driven insights
- Example Initiative: Partnership with **Nuclear Energy Institute** to create curated ADAMS datasets for AI-enabled tools.

Bottom Line: AI is not replacing expertise - it is augmenting NRC staff to improve mission outcomes.

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NRC AI Use Cases

Productivity

- Microsoft 365 Copilot
- OpenAI ChatGPT Enterprise
- Google Gemini for Government

Mission Focused

- SimplifAI
- Microsoft 365 Copilot

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AI Use Case Framework

SCALE Framework for Advancing High Value-Low Risk AI Use Cases

START SMART. FINISH AT SCALE.

S — **S**earch for Opportunities

C — **C**onfirm Enterprise Fit

A — **A**ssemble and Pilot

L — **L**aunch Across Teams and Processes

E — **E**valuate and Evolve

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Questions

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